1.24 Graph Sketching: Exponentials and Logarithms

Worked Solutions

1 a)

$y = 2^x$.

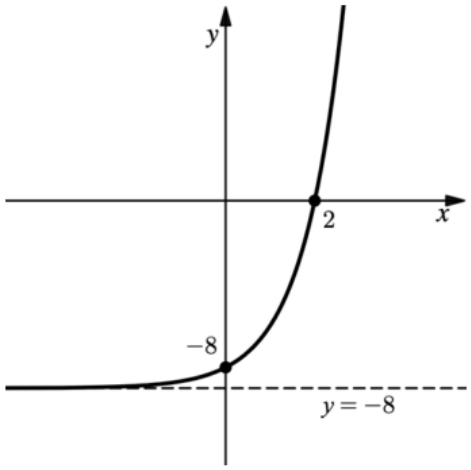
1 b)

$y = 2^x + 2$.

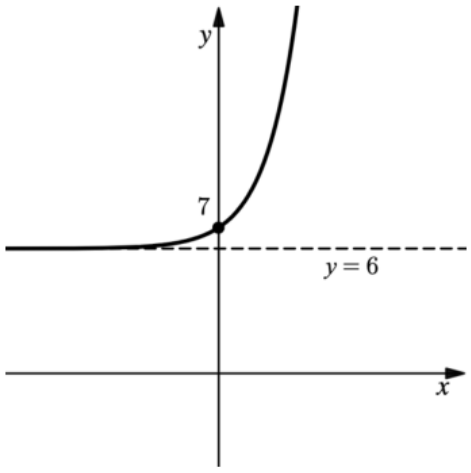
1 c)

$y = 2^x - 8$.

2 a)



2 b)



3 a)

$y = 2^{x-a} + 4$
$(0, \frac{17}{4}) : \frac{17}{4} = 2^{-a} + 4 \Rightarrow 2^{-a} = \frac{1}{4} \Rightarrow 2^{-a} = 2^{-2}$
$\Rightarrow a = 2$
$y = 2^{x-2} + 4.$

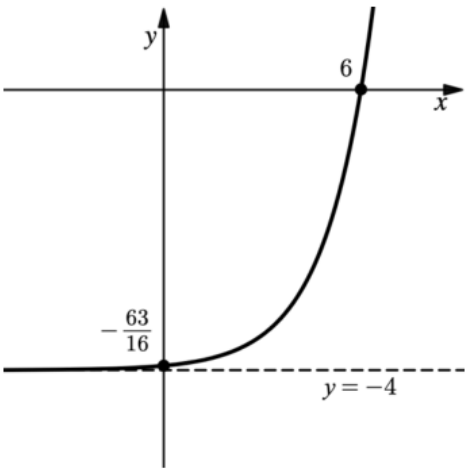
3 b)

$y = 2^{x-a} - 4$
$(0, -2) : -2 = 2^{-a} - 4 \Rightarrow 2^{-a} = 2 \Rightarrow a = -1$
$y = 2^{x+1} - 4.$

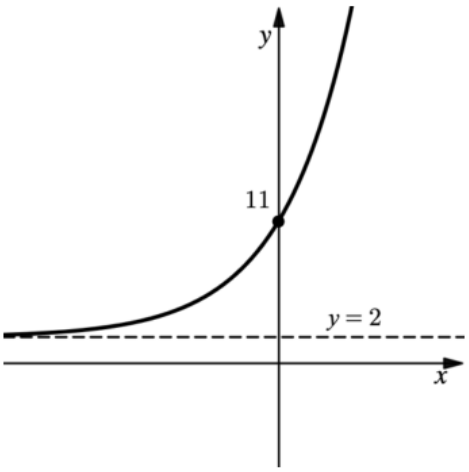
3 c)

$y = 2^{x-a} - 1$
$(0, 7) : 7 = 2^{-a} - 1 \Rightarrow 2^{-a} = 8 \Rightarrow a = -3$
$y = 2^{x+3} - 1.$

4 a)



4 b)



5 a)

$y = 2^{a-x} + 2$
$(0, 6) : 6 = 2^a + 2 \Rightarrow 2^a = 4 \Rightarrow a = 2$
$y = 2^{2-x} + 2.$

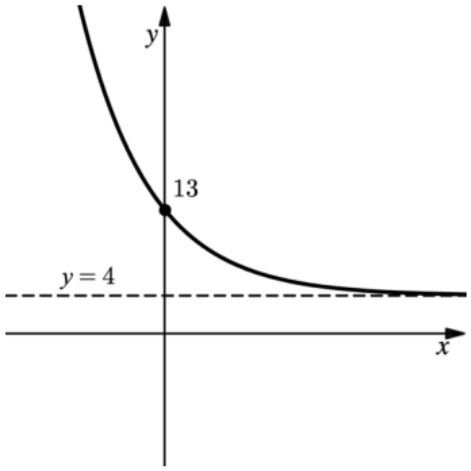
5 b)

$y = 2^{a-x} - 2$
$(0, 6) : 6 = 2^a - 2 \Rightarrow 2^a = 8 \Rightarrow a = 3$
$y = 2^{3-x} - 2.$

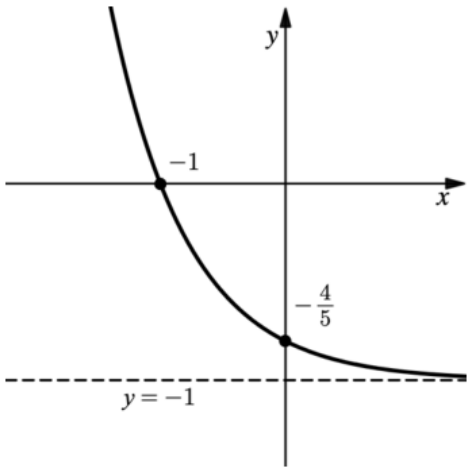
5 c)

$y = 2^{a-x} - 1$
$(0, 15) : 15 = 2^a - 1 \Rightarrow 2^a = 16 \Rightarrow a = 4$
$y = 2^{4-x} - 1.$

6 a)



6 b)



7 a)

$y = 2 - 2^{x-a}$
$(0, \frac{7}{4}) : \frac{7}{4} = 2 - 2^{-a} \Rightarrow 2^{-a} = \frac{1}{4} \Rightarrow 2^{-a} = 2^{-2} \Rightarrow a = 2$
$y = 2 - 2^{x-2}$.

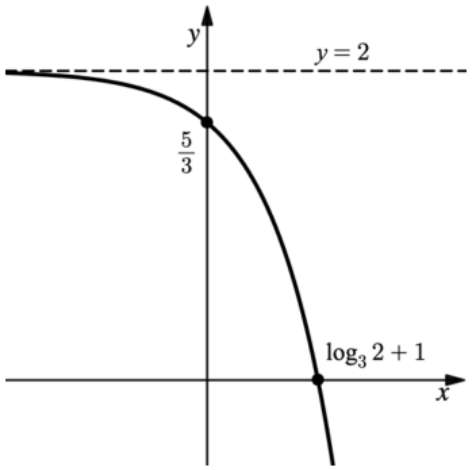
7 b)

$y = -4 - 2^{x-a}$
$(0, -12) : -12 = -4 - 2^{-a} \Rightarrow 2^{-a} = 8 \Rightarrow a = -3$
$y = -4 - 2^{x+3}$.

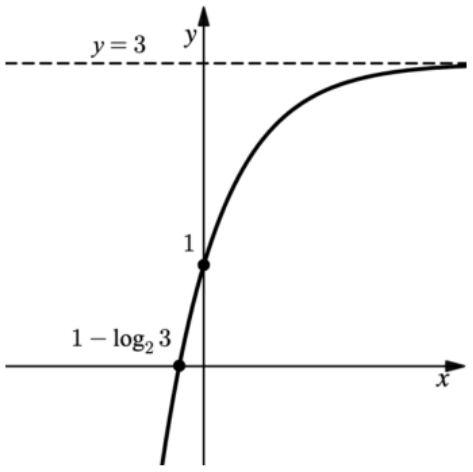
7 c)

$y = 3 - 2^{x-a}$
$(0, 1) : 1 = 3 - 2^{-a} \Rightarrow 2^{-a} = 2 \Rightarrow a = -1$
$y = 3 - 2^{x+1}$.

8 a)



8 b)



9 a)

$a = 2$
$y = \log_2(x - 2)$.

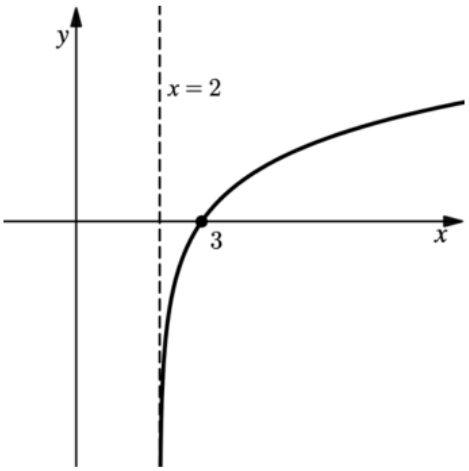
9 b)

$a = -4$
$y = \log_2(x + 4)$.

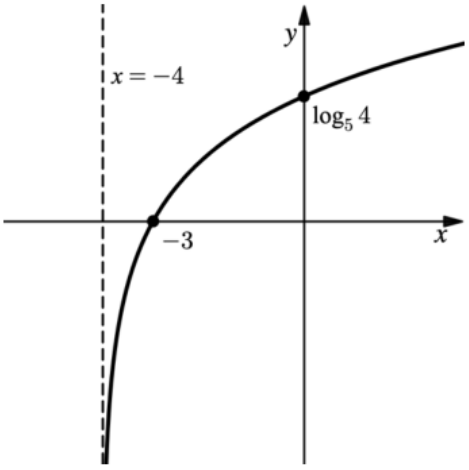
9 c)

$a = 10$
$y = \log_2(x - 10)$.

10 a)



10 b)



11 a)

$y = \log_2 x + b$
$(\frac{1}{2}, 0) : 0 = \log_2 \frac{1}{2} + b \Rightarrow 0 = -1 + b \Rightarrow b = 1$
$y = \log_2 x + 1$.

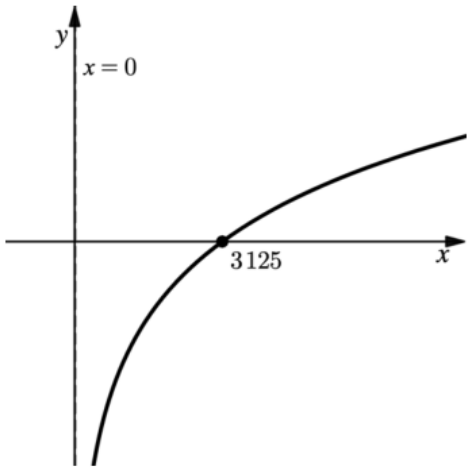
11 b)

$y = \log_2 x + b$
$(4, 0) : 0 = \log_2 4 + b \Rightarrow 0 = 2 + b \Rightarrow b = -2$
$y = \log_2 x - 2.$

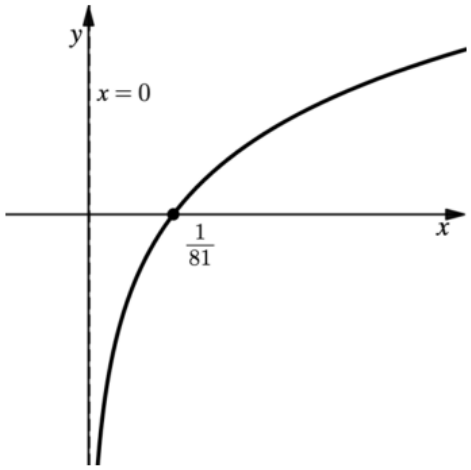
11 c)

$y = \log_2 x + b$
$(\frac{\sqrt{2}}{2}, 0) : 0 = \log_2 \frac{\sqrt{2}}{2} + b \Rightarrow 0 = -\frac{1}{2} + b \Rightarrow b = \frac{1}{2}$
$y = \log_2 x + \frac{1}{2}.$

12 a)



12 b)



13 a)

$y = \log_2(x - 2) + b$
$(4, 0) : 0 = \log_2(2) + b \Rightarrow 0 = 1 + b \Rightarrow b = -1$
$y = \log_2(x - 2) - 1.$

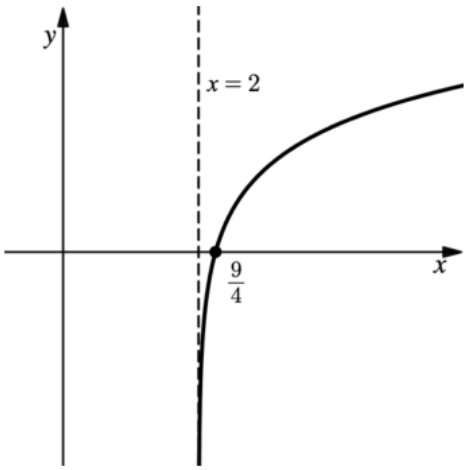
13 b)

$y = \log_2(x+4) + b$
$(0,5) : 5 = \log_2 4 + b \Rightarrow 5 = 2 + b \Rightarrow b = 3$
$y = \log_2(x+4) + 3.$

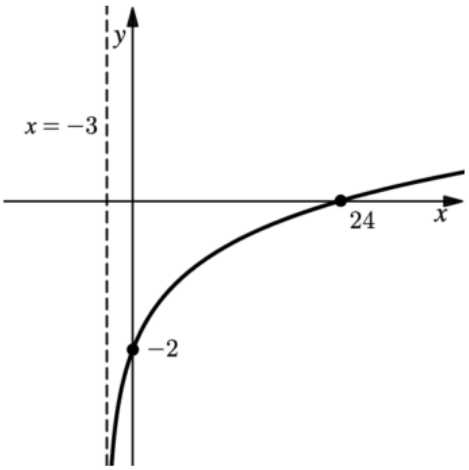
13 c)

$y = \log_2(x+1) + b$
$(0,-3) : -3 = \log_2 1 + b \Rightarrow b = -3$
$y = \log_2(x+1) - 3.$

14 a)



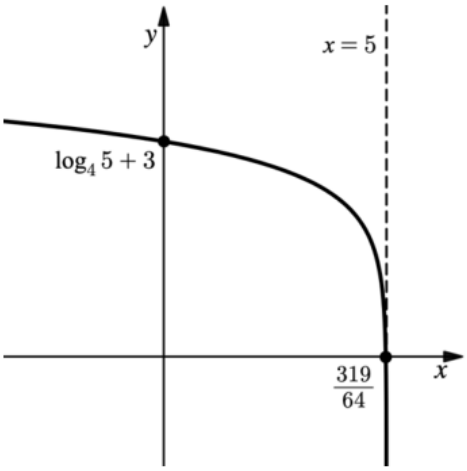
14 b)



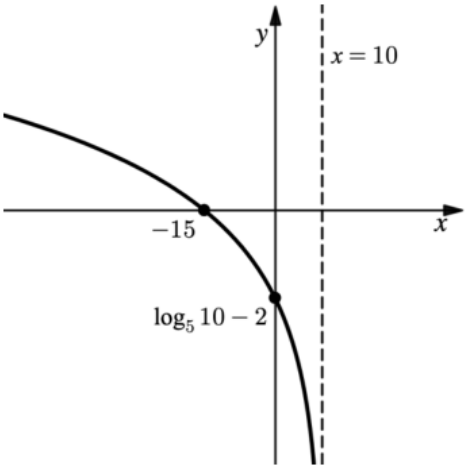
15)

$y = \log_2(4-x) + b$
$(0,5) : 5 = \log_2 4 + b \Rightarrow b = 3$
$y = \log_2(4-x) + 3.$

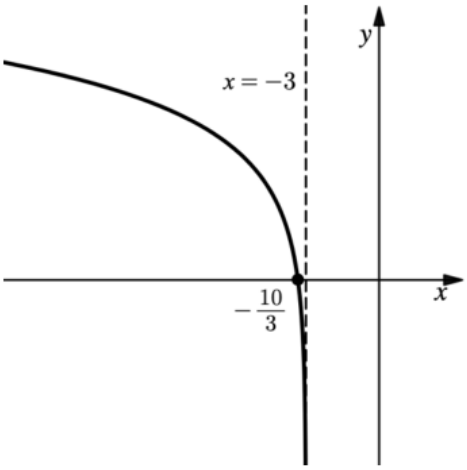
16 a)



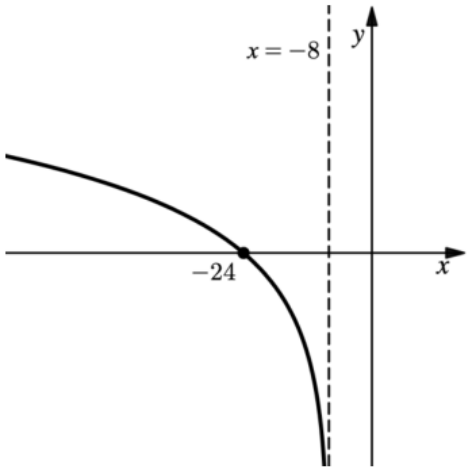
16 b)



16 c)



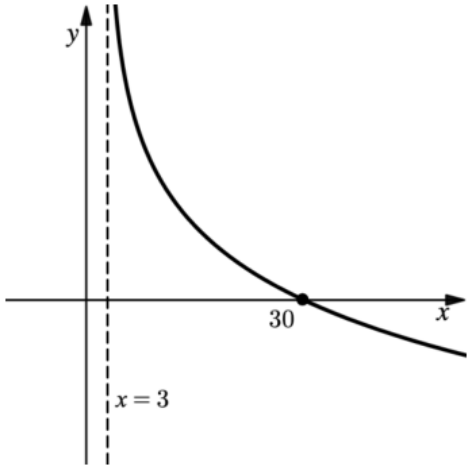
16 d)



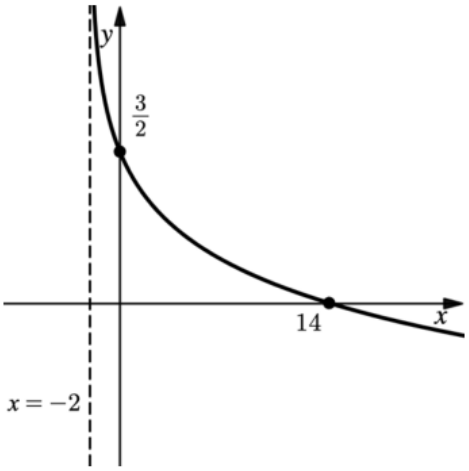
17)

$y = b - \log_2(x-5)$
$(37, 0) : 0 = b - \log_2 32 \Rightarrow b = 5$
$y = 5 - \log_2(x-5)$.

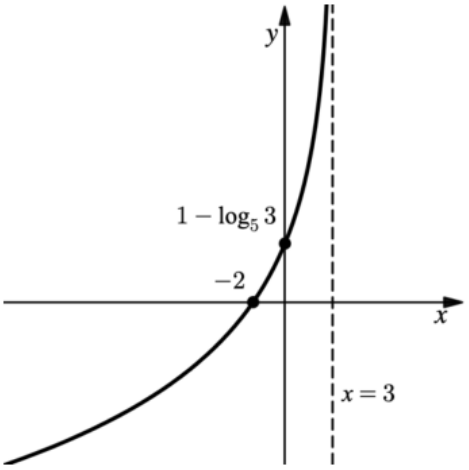
18 a)



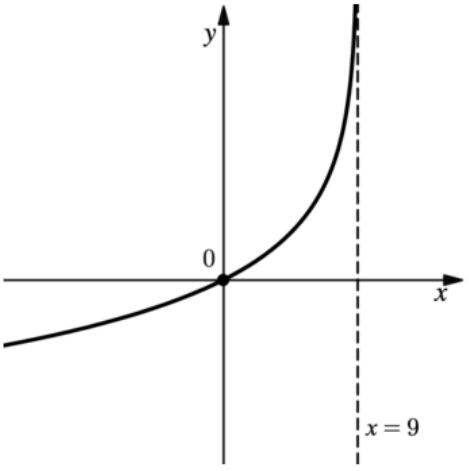
18 b)



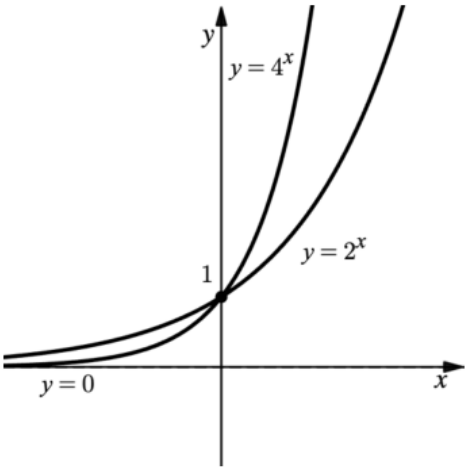
18 c)



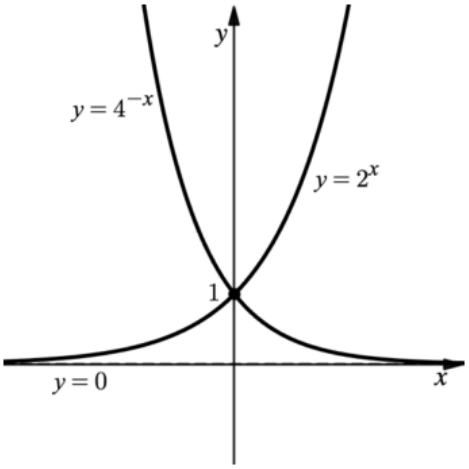
18 d)



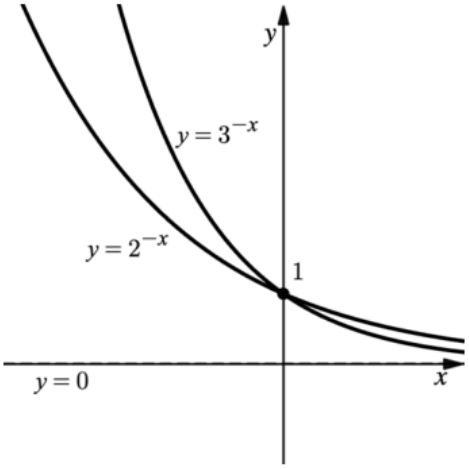
19 a)



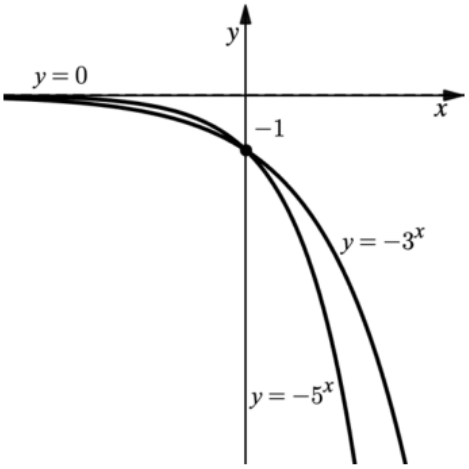
19 b)



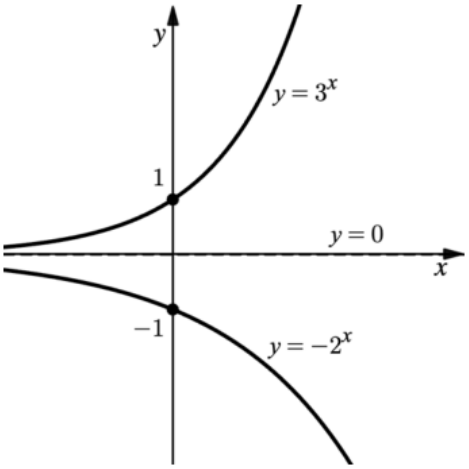
19 c)



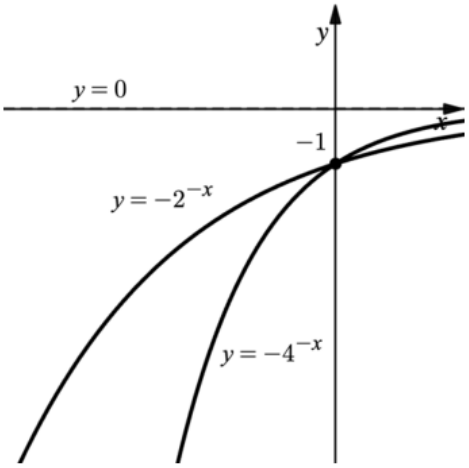
20 a)



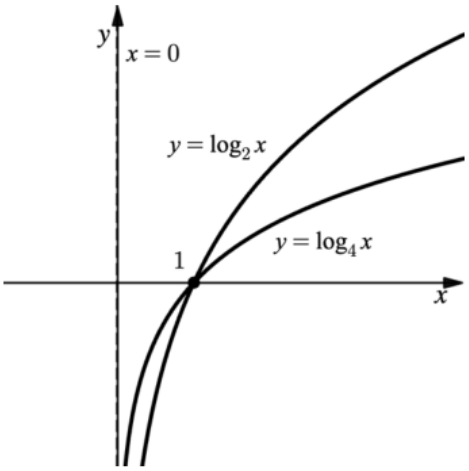
20 b)



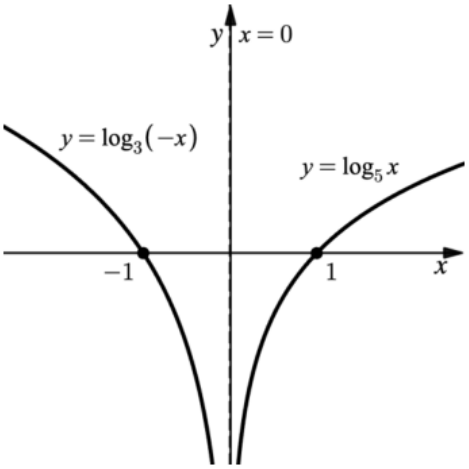
20 c)



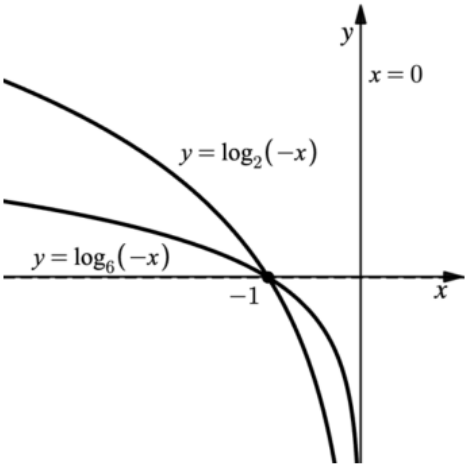
21 a)


y
x = 0
y = log2 x
y = log4 x
1
x

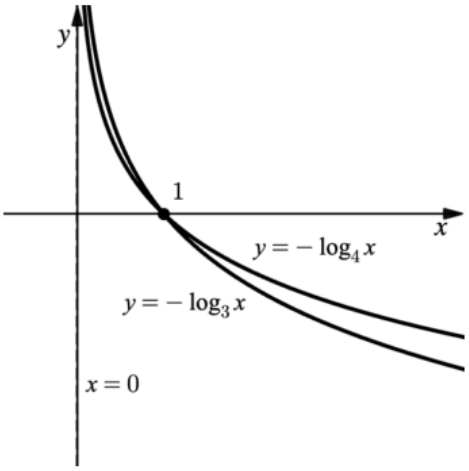
21 b)


y
x = 0
y = log3(−x)
y = log5 x
−1
1
x

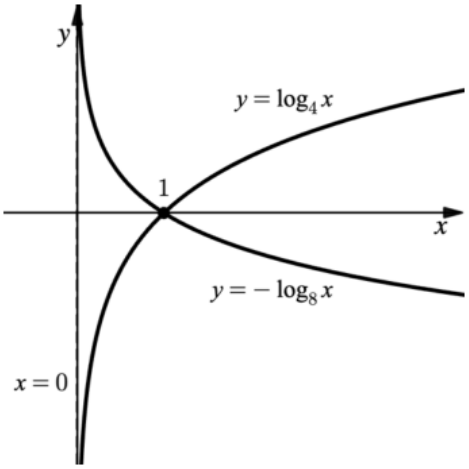
21 c)


y
x = 0
y = log2(−x)
y = log6(−x)
−1
x

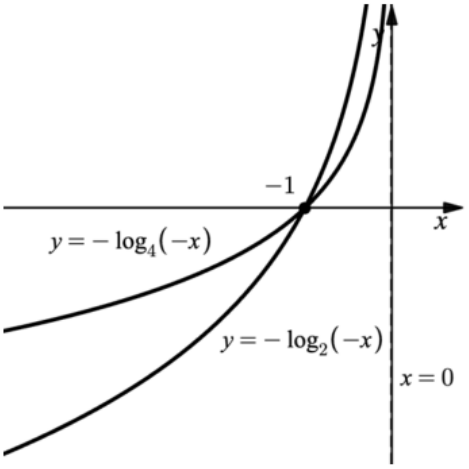
22 a)


y
1
x
$y = -\log_4 x$
$y = -\log_3 x$
$x = 0$

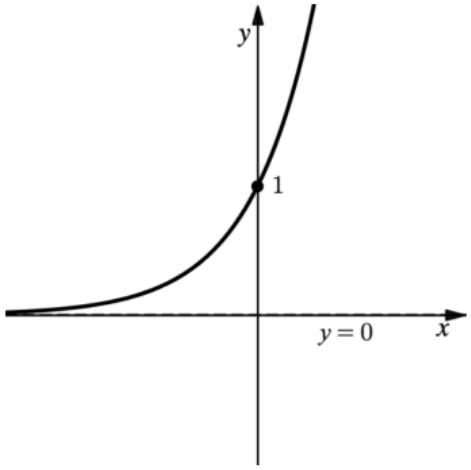
22 b)


y
$y = \log_4 x$
1
x
$y = -\log_8 x$
$x = 0$

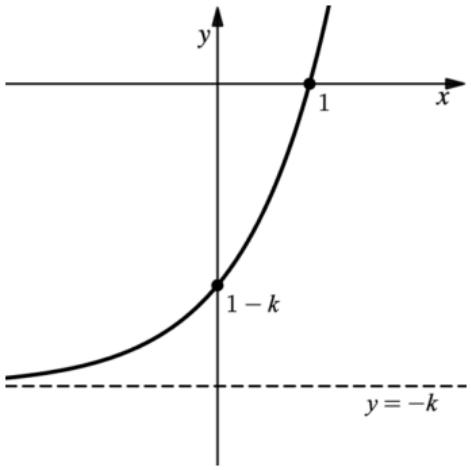
22 c)


y
−1
x
$y = -\log_4(-x)$
$y = -\log_2(-x)$
$x = 0$

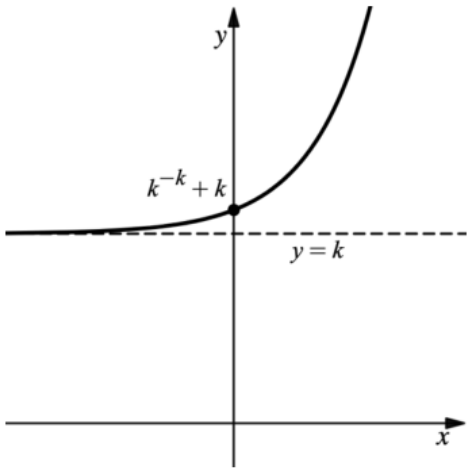
23 a)



23 b)



23 c)



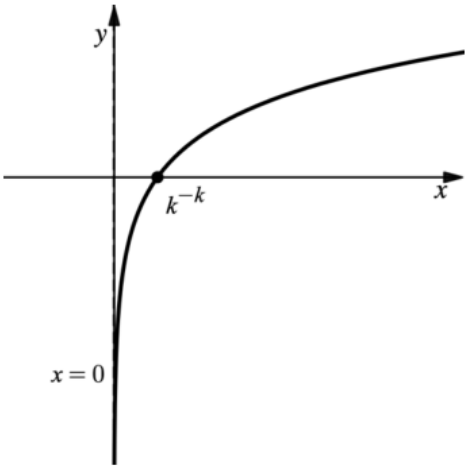
24 a)

$y = k^{k-x} - k$.

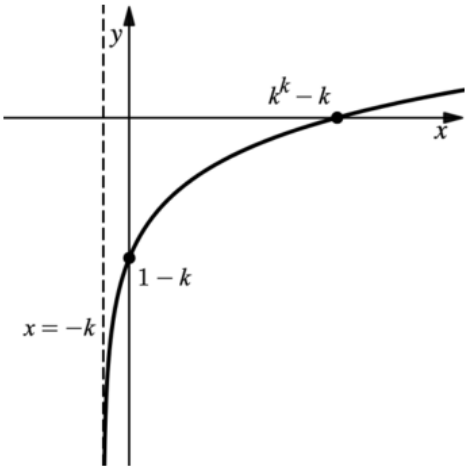
24 b)

$y = 2k - k^{k-x}$.

25 a)



25 b)



26 a)

$y = k - \log_k(x + k)$.

26 b)

$y = k - \log_k(-k - x)$.